

Denison University
Data Analytics Program

A Decade of Decline: Statistical Modeling of Bank of America's Physical Branch Closures

DA 352 - Advanced Predictive Methods
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1 Introduction

1.1 Background and Objective

In response to growing digital adoption and shifting customer preferences, many U.S. banks have reduced their physical presence by closing branches. This project focuses on Bank of America and seeks to predict when and whether a branch will close, using a ten-year panel dataset. By assessing factors such as financial performance, local competition, and broader socioeconomic conditions, we aim to understand what drives branch survival in an evolving banking landscape.

The main question of our research is: *What are the most important predictors of branch closure?* To address this, we applied various statistical methods that were covered in the course. Classification techniques such as decision trees and boosting models are used to predict branch closures and assess feature importance. In addition, to predict how long until closure and the relative impact of different predictors on branch survival, we used survival analysis, including Kaplan-Meier survival curves and the Cox Proportional Hazards model. Together, these methods help uncover the patterns and risk factors that drive branch longevity.

1.2 Data Source

The primary dataset used in this project is the Summary of Deposits (SOD), published annually by the Federal Deposit Insurance Corporation (FDIC). This dataset provides detailed information on bank branches across the United States, including geographic coordinates, deposit amounts, and institutional identifiers (Table 1). To enhance the analysis with socioeconomic context, we also incorporated U.S. Census data on estimated household income at the ZIP code level.

2 Ethical Consideration

This analysis is conducted mainly for educational purposes as part of an academic project. All data used in the study is publicly available and sourced from reputable institutions, including the FDIC and the U.S. Census Bureau. No proprietary, sensitive, or personally identifiable information is used. While the primary stakeholders are classroom personnel (students and instructors), the findings may also be of interest to banking professionals, such as branch managers, analysts, and strategic planners, who are concerned with operational efficiency and resource allocation. We aim for this analysis to not only fulfill academic objectives but also provide insights that could inform discussions on branch sustainability and resource allocation in the banking sector. Ethical care has been taken to ensure that the analysis is unbiased, respects community impact, and avoids misinterpretation or misuse of the data.

3 Data Exploration

3.1 Data Cleaning and Processing

Table 1: Variables Definitions

Variables	Definition
year	Year the observation was recorded (2014-2024)
bank_name	Name of the bank
branch_id	Unique identifier for the branch
state_abb	State abbreviation
city	City where the branch is located
county	County where the branch is located
combined_stat_area	Combined statistical area code
deposits	Total deposits at the branch
region_code	Indicator: 1 - rural, 2 - micropolitan, 3 - metropolitan
branch_name	Name of the branch
established_date	Date the branch was established
latitude	Latitude coordinate of the branch
longitude	Longitude coordinate of the branch
state_name	Full state name
total_nearby_competitor_branches	Total number of competing branches nearby
nearby_boa_branches	Number of nearby Bank of America branches
total_nearby_boa_deposits	Deposits held in nearby Bank of America branches
total_nearby_competitor_deposits	Deposits held in nearby competing branches
branch_share	Share of deposits held by this branch in its market
market_share	Share of deposits held by this bank in its market
network_equity	Measure of the network value of the branch (Market Share divided by Branch Share)
deposit_change	1 year change in deposits
closed	Indicator: 1 if the branch is closed, 0 if still open
zipcode	Branch ZIP code
estimate_household_median_income	Estimated median income in the branch's area
estimate_household_mean_income	Estimated mean income in the branch's area
entry_time	Maximum of 0 and (2014-established_year), representing the delayed entry or left truncation
exit_time	The time since opening until 2024 or until closure

To construct a panel of branch activity over time, we collected and merged 10 annual FDIC SOD reports from 2014 to 2024. A branch was marked as closed if it appeared in the previous year's report but was absent in the current year's report. For such branches, we retained the most recent available record prior to closure. Conversely, branches that appeared in the 2024 report were considered active, and their most recent data was taken from that year. We also merged the SOD data with Census household income data using ZIP codes. This allowed us to assign each branch an estimated median and mean household income, capturing the economic environment surrounding each location. Because the number of rows with missing values was minimal (fewer than 40), we decided to remove them from the analysis for simplicity and to maintain data integrity.

In addition to internal financial and external economic factors, we examined the influence of branch density and competition within local markets. For each branch, we computed the number of nearby Bank of America branches and competing branches located within a 10-mile radius. Using this information, we calculate branch share, market share, and network equity (Table 1). These features will allow us to assess whether the branch is currently in good condition or not.

Finally, for survival analysis purposes, we structured the data set to correctly define time-at-risk and censoring. We first convert each branch’s establishment date into a date format and extract the year of opening. Since the study window begins in 2014, we defined the entry time as the maximum of 0 and (2014 minus the establishment year), representing delayed entry; only branches active in 2014 enter the risk set. The exit time was defined as the number of years from opening to 2024 or until the branch was closed. Branches with exit times less than or equal to entry times were excluded to ensure valid survival durations. These transformations allow us to perform left-truncated survival analysis, which accounts for the fact that branches had to survive until 2014 to be observed. By doing so, we want to prevent time bias, which could lead to incorrect survival estimates.

3.2 Exploratory Data Analysis

The final cleaned dataset contains 5,412 observations of Bank of America branches across the United States, with closure status as the primary outcome of interest. Of these, 3,738 branches were still operating as of 2024, while 1,674 branches had closed during the study period.

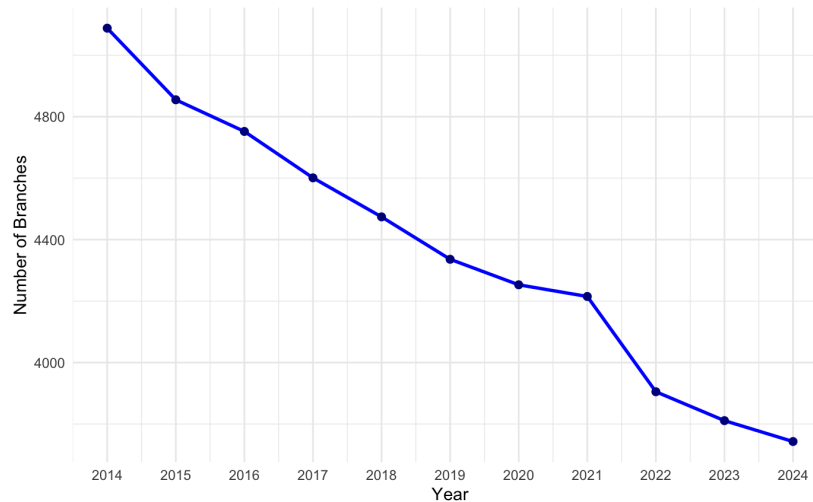


Figure 1: Bank of America Branch Count (2014–2024)

First, to understand the characteristics and patterns associated with branch closure, we first examined the number of BoA’s branches from 2014 to 2024. In Figure 1, the number of Bank of America branches has steadily decreased in the past decade, possibly due to the broader industry shift towards digital banking and cost efficiency. In 2021, a noticeably sharper decline occurs, likely corresponding to the operational impacts of the COVID-19 pandemic, including accelerated closures and reduced in-person banking demand.

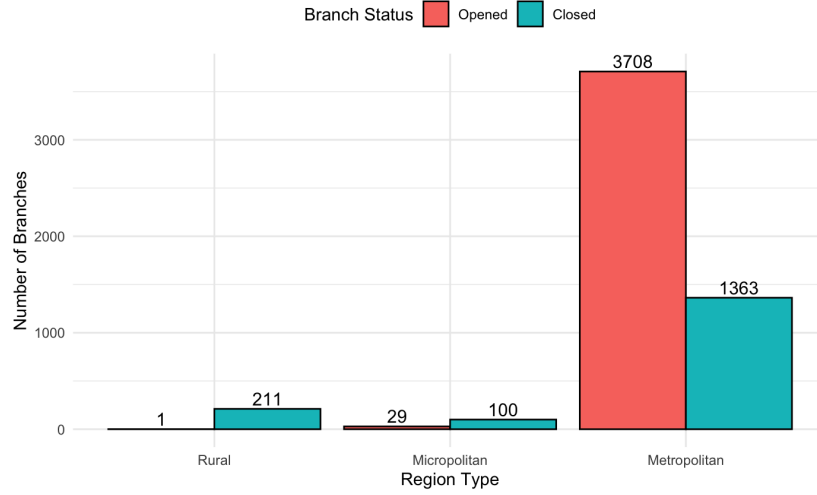


Figure 2: Branch Status by Region Type

To go more in-depth, we examine branch status by region type to uncover geographic patterns behind closures and openings. At first glance, we observe that Bank of America’s recent trend over the past ten years has been to open new branches primarily in metropolitan areas (Figure 2). This is unsurprising given the higher population density and greater demand for financial services in these regions. In contrast, closures are more common in rural and micropolitan areas, where branch openings are minimal or non-existent. This suggests a strategic consolidation of physical infrastructure toward urban markets with higher economic activity. It is also worth mentioning that the dataset is imbalanced, with approximately 70% of branches remaining open and only 30% closing, which has important implications for model training and evaluation in the next stages of analysis.

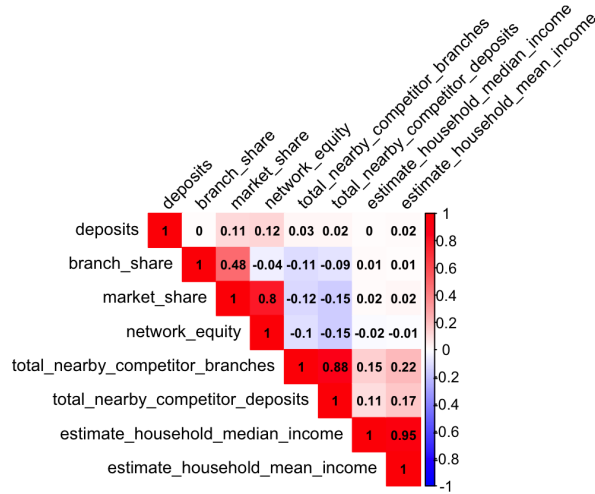


Figure 3: Correlation Matrix of Key Predictors

To identify possible multicollinearity, which can affect model performance, we created a correlation matrix of all predictors (Figure 3). As expected, median and mean household income are highly correlated ($r = 0.95$), mainly because they are coming from the same local census data. Similarly, market share and network equity show strong correlation ($r = 0.80$), since network equity is a function of market share. Total nearby competitor branches and competitor deposits are also closely related ($r = 0.87$), reflecting the natural link between branch presence and deposit volume in an area.

So far, our exploratory data analysis has revealed clear trends in branch closure patterns over the past decade, with an overall decline in branch count, particularly in rural areas, and strong associations between closure and variables like deposit performance, regional type, and household income. It is also noted that several pairs of highly related predictors could influence model behavior. These initial insights provide valuable context and direction for deeper statistical analysis, which we will perform in the next section.

4 Statistical Analysis and Interpretation

To analyze branch closure patterns, we apply two main statistical approaches: supervised learning using tree-based methods and survival analysis techniques. The processed dataset was split into 70% training and 30% testing sets to evaluate model performance and generalizability.

4.1 Supervised Method: Tree-based Model

Supervised learning models are trained on labeled data to predict outcomes based on input features. Therefore, we use all the computed features to build a model that predicts whether a branch will close. We employ two tree-based methods, decision trees and boosting, which are well-suited for this task since we care about interpretability, allowing us to identify the most influential factors driving branch closure. In addition, these methods, especially boosting, are well-known for their ability to handle missing values, highly correlated variables, as well as imbalanced data. Therefore, given the nature of our dataset, we believe tree method is the correct choice for us.

4.1.1 Decision Tree

We started with the most basic tree structure, which is Decision Tree. Given its high interpretability, it allows us to clearly see how different features contribute to predicting branch closure. However, decision trees can also be prone to overfitting, especially on complex or noisy datasets, which is the case here given the number of features and data imbalance. Despite this limitation, the decision tree provides a useful baseline and helps us gain initial insights into important predictors.

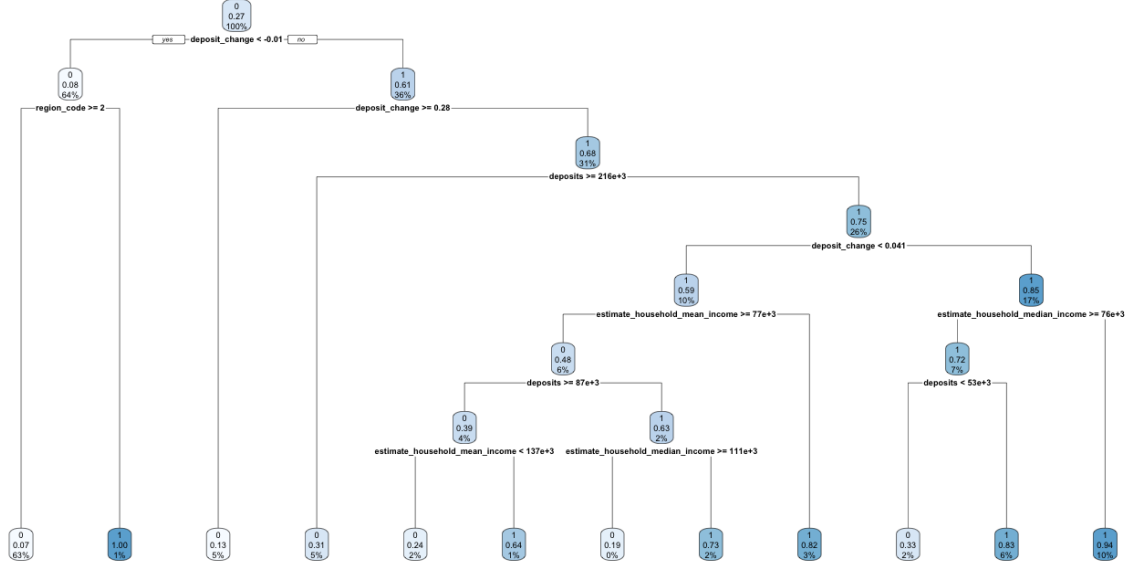


Figure 4: Decision Tree for Branch Closure

The tree structure from Figure 4 reveals that the most influential factor in predicting branch closure is deposit change due to its first split, followed by deposit amount, household income levels, and region type. For instance, branches experiencing a significant drop in deposits (e.g., deposit change < -0.1) and located in less urban areas are more likely to close. On the other hand, higher deposits and higher surrounding income levels are associated with branches remaining open. The tree’s structure confirms that financial performance and local socioeconomic conditions are key signals of branch survival.

Table 2: Confusion Matrix for Decision Tree

	Actual 1 (Closed)	Actual 0 (Opened)
Predicted 1 (Closed)	273 (TP)	60 (FP)
Predicted 0 (Opened)	127 (FN)	1032 (TN)

From the confusion matrix we obtained by using the tree to predict on the testing data, we achieved an overall accuracy of 87.5%, meaning it correctly predicts the branch status for 87.5% of the test data (Table 2). Among actual open branches, 94.51% were correctly identified. Yet, regarding the actual closed branches, only 68.5% were captured, showing that the model still misses about 32%. The balanced accuracy of 81.4% adjusts for this class imbalance by averaging sensitivity and specificity, giving a more fair assessment of performance across both outcomes.

This result gives us a moderate performance in predicting branch closure. However, some of the closed branches are still misclassified, as well as the decision of the model are largely depended on deposit and deposit growth. Therefore, this motivates us to shift to more advanced methods like boosting, which we are going to perform next, for improved performance.

4.1.2 Boosting

Boosting is an approach for enhancing the predictions resulting from a decision tree. In simpler terms, boosting works in a way that the trees are grown *sequentially*: each tree is grown using information from previously grown trees (James et al. 2013). We then implemented boosting using XGBoost in R, which stands for Extreme Gradient Boosting. This method uses a gradient descent algorithm to minimize the loss when adding new models (IBM Developer). We applied 10-fold cross-validation to fine-tune some key hyperparameters in the model, which include: number of boosting iterations, the learning rate, the maximum depth of the tree, and the minimum loss reduction required to make further partition of leaf node of the tree. We prioritized cross-entropy loss as the evaluation metric during tuning process and trained the final XGBoost model using the entire training set. Then we evaluated the model’s predictions on the test set using the optimal classification threshold determined by the Youden Index from the ROC curve (RDocumentation). This approach ensures a balanced trade-off between sensitivity and specificity by selecting the threshold that maximizes the distance from the chance line.

Table 3: Confusion Matrix for XGBoost

	Actual 1 (Closed)	Actual 0 (Opened)
Predicted 1 (Closed)	353 (TP)	119 (FP)
Predicted 0 (Opened)	47 (FN)	973 (TN)

Looking at the confusion matrix derived from predicting on the testing set, we can see that the model demonstrates strong overall performance, achieving an accuracy of 88.87%. More importantly, it balances the ability to detect both classes effectively, as shown by its sensitivity of 88.25% and specificity of 89.10%. This means the model correctly identifies 88.25% of actual closed branches and 89.10% of actual open branches, indicating a reliable classification. Overall, these findings indicate that fine-tuning via cross-validation improved the model’s generalizability by lowering overfitting and enhancing its capacity to detect branches at danger of closure.

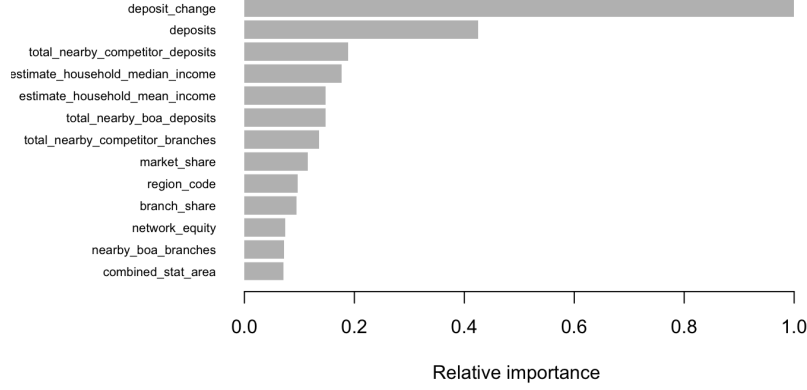


Figure 5: Feature Importance (XGBoost Model)

The feature importance plot from the final XGBoost model further confirms the dominant role of deposit-related variables in predicting branch closure (Figure 5). Specifically, change in deposit levels over time is by far the most influential predictor, followed by total deposit amount. These findings suggest that both short-term financial performance and overall deposit volume are critical indicators of a branch’s viability. Socioeconomic factors such as mean and median household income also appear prominently, indicating the impact of local economic conditions on closure risk. While other variables, such as nearby competitor activity and market share, are included in the model, their relative importance is much lower, reinforcing the centrality of direct financial and demographic measures in forecasting branch outcomes.

4.1.3 Model Comparison

Table 4: Comparison of Model Performance Metrics

Model	Accuracy	Sensitivity	Specificity
Decision Tree	0.875	0.682	0.945
Boosting (XGBoost)	0.888	0.882	0.891

Table 4 presents a comparison between the Decision Tree and Boosting (XGBoost) models across key performance metrics. While both models perform well, the accuracy for Boosting is slightly higher than Decision Tree. Notably, Boosting outperforms the Decision Tree in terms of sensitivity (0.882 vs. 0.682). This suggests that XGBoost is significantly better at correctly identifying closed branches, which is crucial if detecting closures is a priority. Although the Decision Tree has a higher specificity (0.945 vs. 0.891), though indicates that it is slightly better at identifying open branches, comes at the cost of missing more actual closures. Overall, even when decision tree offers strong interpretability and a clear visual explanation of how features like deposit change and household income influence closure decisions, we would say that XGBoost is the more effective model for predicting branch closures in this analysis.

4.2 Survival Analysis

For the second part of our analysis, we shift from classification to survival analysis to examine the timing and risk of branch closure over time. This approach allows us to model not just whether a branch closes, but when closure occurs, providing a better understanding of branch longevity. We employ three key methods: the Kaplan-Meier survival curve to estimate survival probabilities over time, the log-rank test to compare survival distributions across groups, and the Cox Proportional Hazards model to quantify how specific factors affect the hazard, or risk, of closure.

4.2.1 Kaplan-Meier Survival Curves

The Kaplan-Meier Survival Curves uses the survival function to quantify the probability of surviving past time (James et al 2013). We first estimated an overall survival curve using the Kaplan-Meier estimator, incorporating left-truncation to account for the fact that our observation window begins in 2014. Left-truncation ensures that only branches already in operation by 2014 and still at risk of closure during the study period are included in the analysis, thereby avoiding bias from including branches that had already closed before data collection began.

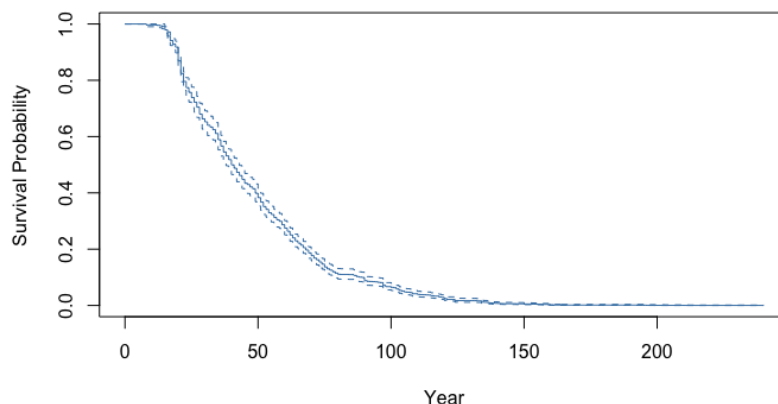


Figure 6: Kaplan-Meier Survival Curve (Left-Truncated)

The resulting survival curve (Figure 6) shows the probability that a branch remains open as a function of time since branch establishment. Each drop in the curve corresponds to a closure event, while the overall downward trend reflects the cumulative risk of closure over time. The horizontal axis represents the number of years since the opening of a branch, and the vertical axis indicates the proportion of branches that have survived (i.e., remained open) up to that point. The shaded gray area around the curve represents the 95% confidence interval, showing the range within which the true survival probability will likely fall. It can be seen that there is a steep decline in the early years, suggesting a high risk of closure shortly after establishment, with survival probability dropping below 50% by around year 50.

4.2.2 Checking the Proportional Hazards Assumptions

Next, we want to use the Cox Proportional Hazards (PH) model, a widely used method in survival analysis that estimates the survival time as a function of the covariates (James et al. 2013). This model requires us to

make the following assumptions (Wilson 2018):

- Independent observations: Each branch is assumed to be conditionally independent, though shared regional effects may introduce unobserved clustering.
- Non-informative censoring: We assume the reason a branch remains open in 2024 is unrelated to unobserved closure risk. This is untestable but partially mitigated by including key predictors.
- Proportional hazards: The survival curves for two different strata of a risk factor must have hazard functions that are proportional over time.

Ensuring that these assumptions hold is important because violations can lead to biased estimates and incorrect conclusions about the effect of covariates on survival outcomes.

Table 5: Assumptions Check (cox.zph)

	chisq	df	p-value
deposits	55.44	1	9.6e-14
deposit_change	101.17	1	< 2e-16
network_equity	2.55	1	0.11
estimate_household_median_income	1.84	1	0.17
region_code	2.00	1	0.16

Table 5 presents the results of the proportional hazards assumption test using Schoenfeld residuals (Kassambara). On a variable level, both deposits ($p < 0.05$) and deposit growth ($p < 0.05$) violate the PH assumption, suggesting that their effects on the hazard of branch closure vary over time. On the other hand, variables such as Network Equity, Estimate Household Median Income, and Region Code do not show significant deviations from proportionality (p-values > 0.1), indicating that their effects may be appropriately modeled under the PH assumption. To address the violations, time-varying effects were later introduced for the non-proportional variables.

4.2.3 Time-Dependent Cox Proportional Hazards Model

To handle violations discussed in the previous part, we introduced time-interaction terms, in which we apply deposit and deposit growth with a function of time like $\log(t)$. This will allow the deposit effects to vary with log-transformed time (Dunkler et al). We then used some of the most important features identified by the tree-based models above to fit the Cox PH model. In addition, even though some features are identified as important by the tree-based model, we decided to remove them to avoid multicollinearity issues that might occur.

Table 6: Summary of Cox PH Model Coefficients

	Coefficient	Std. Error	z-value	p-value
tt(deposits)	-2.521e-08	1.375e-08	-1.834	0.0667
tt(deposit_change)	2.922e-03	2.460e-03	1.188	0.2349
network_equity	-8.156e-02	3.326e-02	-2.452	0.0142*
estimate_household_median_income	-1.034e-05	8.942e-07	-11.568	<2e-16***
region_code	-5.010e-01	4.523e-02	-11.075	<2e-16***

Note: Significance codes: *** $p < 0.001$, * $p < 0.05$
Concordance = 0.651

The results (Table 6) show that the time-varying effect of deposits is negative, suggesting that as time progresses, higher deposit levels are associated with a decreasing hazard of closure. However, this effect is not statistically significant ($p > 0.05$). Similarly, deposit growth shows a positive time-varying effect, implying an increasing hazard over time, but this also lacks statistical significance with high p-value. On the other hand, Network Equity has a significant negative effect ($p < 0.05$), indicating that branches with more competitive positioning face a lower risk of closure. Both Estimate Household Median Income and Region Code are also significant with low p-value and negative coefficients, suggesting that branches located in wealthier regions or certain geographic areas have significantly better survival prospects. The concordance index of 0.651 indicates moderate predictive accuracy of the model. In other words, this means the model correctly ranks the relative risk of closure between two randomly chosen branches about 65% of the time (James et al. 2013).

4.2.4 Model Validation: Cross-Validation

To evaluate the generalizability of the Cox model, we conducted a 5-fold cross-validation method, which includes partitioning the data into five equal subsets, iteratively training the model on four subsets while testing on the remaining one. By averaging the performance across all five folds, we gain a more robust estimate of how the model is expected to perform on unseen data, mitigating the risk of overfitting that can occur when evaluating model performance on the training set alone (Simon et al). The resulting average concordance index (C-index) was 0.6664, meaning the model correctly ranks branch closure risk approximately 67% of the time. While not perfect, the result indicates that the model is reasonably effective at identifying higher-risk branches, providing a practical tool for supporting strategic decisions in branch network management.

5 Conclusions

5.1 Summary of Findings

In conclusion, this study used supervised machine learning techniques, specifically decision trees and boosting, to predict Bank of America branch closures with a focus on interpretability and accuracy. The decision tree model provided a clear baseline by revealing that deposit change, total deposits, household income, and region type are key predictors of closure. Although the model performed well overall, it had limited sensitivity to actual closures due to class imbalance. Boosting, implemented through XGBoost, addressed these limitations by improving both sensitivity and overall accuracy. Through cross-validation and threshold optimization, the model achieved nearly 89% accuracy and proved effective in identifying high-risk branches based on financial and demographic indicators.

To understand the timing of branch closures, we incorporated survival analysis using Kaplan-Meier estimators and Cox Proportional Hazards models. The Kaplan-Meier curves showed that the highest risk of closure tends to occur in the early years of a branch’s operation. The Cox models allowed us to assess the influence of specific predictors on closure risk over time, with network equity, regional income, and region code emerging as significant factors. Time-dependent modeling was used to address violations of the proportional hazards assumption and improve accuracy. While the final survival model achieved a moderate concordance index of 0.66, it still offered valuable insights into long-term branch viability and can help inform strategies for branch network management.

5.2 Limitations

While our analysis provides meaningful insights into the factors driving Bank of America branch closures, several limitations should be acknowledged. First, external events such as the COVID-19 pandemic may have had

an excessive impact on branch closures during the observed period. For example, in 2021, we observed a sharp decline in the number of branches, which may be less about underlying financial performance and more about short-term, crisis-driven decisions. Such disruptions introduce noise into the data and can distort long-term patterns of closure risk. Additionally, a branch can be closed even if it is performing well, simply because the bank chooses to consolidate locations or shift toward digital services as part of a broader resource optimization strategy. These strategic decisions may not be captured by the variables included in our models.

Second, our survival analysis is based on a ten-year window from 2014 to 2024. As a result, the models only capture the closure risk for branches that existed and were observed during this timeframe. Branches established before or after this window may not follow the same risk patterns, limiting the model’s applicability for long-term forecasting. Finally, while we employed widely used modeling techniques such as decision trees, XGBoost, and Cox regression, the implementation and tuning of these models were constrained by the scope of our coursework. Due to limited time spent in class, our models may not be fully fine-tuned for peak performance, and further refinement could enhance predictive accuracy and robustness.

5.3 Future Works

Future research can address these limitations by incorporating internal operational data, refining feature selection, and exploring additional modeling techniques. Relying solely on public data may overlook important branch-level dynamics; therefore, access to internal metrics such as the number of accounts opened, service types, customer demographics, and transaction activity could provide a more complete picture of branch performance. This would help distinguish closures driven by poor performance from those made for strategic resource optimization.

Enhancing predictive accuracy could also be achieved by applying alternative supervised learning methods, such as neural networks or ensemble techniques. These models may better capture complex relationships in the data and improve generalizability across diverse branch types and market conditions. Furthermore, extending the survival analysis framework by incorporating macroeconomic indicators or time-varying external events, such as the COVID-19 pandemic, may offer a better understanding of closure timing and risk. Together, these improvements would deepen insight into branch viability and support more informed decisions in branch network management.

6 References

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7 Code Appendix

Here is the code we used for our analysis:

```
# import packages
library(lubridate)
library(tidyverse)
library(caret)
library(dplyr)
library(ggthemes)
library(glmnet)
library(MASS)
library(class)
library(pROC)
library(randomForest)
library(ggplot2)
library(corrplot)
library(GGally)
library(rpart)
library(rpart.plot)
library(xgboost)
library(gbm)
library(doParallel)
library(survival)
library(survminer)
boa_data <- read.csv("data/cleaned_data/cleaned_data.csv")
boa_data$closed <- as.factor(boa_data$closed)

# Recode region: 1 - rural, 2 - micropolitan, 3 - metropolitan
boa_data <- boa_data %>%
  mutate(region_code = case_when(
    is_metropolitan == 1 ~ 3,
    is_micropolitan == 1 ~ 2,
    TRUE ~ 1
  ))

# Number of BoA bank overtime
branches_year <- c(5088, 4855, 4752, 4601,
                  4474, 4336, 4253, 4215,
                  3905, 3811, 3743)

year <- 2014:2024

branch_data <- data.frame(year, branches_year)

ggplot(branch_data, aes(x = year, y = branches_year)) +
  geom_line(color = "blue", size = 1) +
  geom_point(color = "darkblue", size = 2) +
  labs(x = "Year",
       y = "Number of Branches") +
```



```

scale_x_continuous(breaks = 2014:2024) +
theme_minimal()

# Figure 1
boa_data %>%
  mutate(
    region = recode(factor(region_code),
                        `1` = "Rural",
                        `2` = "Micropolitan",
                        `3` = "Metropolitan"),
    branch_status = recode(factor(closed),
                           `0` = "Opened",
                           `1` = "Closed")
  ) %>%
  group_by(region, branch_status) %>%
  summarise(count = n(), .groups = "drop") %>%
  ggplot(aes(x = region, y = count, fill = branch_status)) +
  geom_bar(
    stat = "identity",
    position = position_dodge(width = 0.9),
    color = "black") +
  geom_text(aes(label = count),
            position = position_dodge(width = 0.9),
            vjust = -0.3, size = 4) +
  labs(
    x = "Region Type",
    y = "Number of Branches",
    fill = "Branch Status"
  ) +
  theme_minimal() +
  theme(
    legend.position = "top",
    plot.title = element_text(size = 14, face = "bold")
  )

# Clean the deposit change column
boa_data$deposit_change <- as.numeric(boa_data$deposit_change)
boa_data$deposit_change[boa_data$deposit_change %in% c(0, -1, Inf, -Inf)] <- NA

df_cleaned <- boa_data %>%
  dplyr::select(closed,
                combined_stat_area,
                deposits,
                branch_share,
                market_share,
                network_equity,
                deposit_change,
                nearby_boa_branches,
                total_nearby_competitor_branches,

```

```

        total_nearby_competitor_deposits ,
        total_nearby_boa_deposits ,
        estimate_household_median_income ,
        estimate_household_mean_income ,
        region_code) %>%
filter(if_all(everything(), ~ !is.na(.) & !is.infinite(.)))

# Select only numeric columns for correlation
numeric_vars <- df_cleaned %>%
  dplyr::select(deposits ,
                branch_share ,
                market_share ,
                network_equity ,
                total_nearby_competitor_branches ,
                total_nearby_competitor_deposits ,
                estimate_household_median_income ,
                estimate_household_mean_income)

cor_matrix <- cor(numeric_vars , use = "complete.obs")

corrplot(cor_matrix ,
          method = "color" ,
          type = "upper" ,
          addCoef.col = "black" ,
          number.cex = 0.7 ,
          tl.col = "black" ,
          tl.srt = 45 ,
          tl.cex = 0.9 ,
          col = colorRampPalette(c("blue" , "white" , "red"))(200) ,
          mar = c(0 , 0 , 1 , 0)
)

# Train test split
set.seed(123)
train_index <- createDataPartition(df_cleaned$closed , p = 0.7 , list = FALSE)
train_data <- df_cleaned[train_index , ]
test_data <- df_cleaned[-train_index , ]

# Fit decision tree
tree_model <- rpart(closed ~ . ,
                    data = train_data ,
                    method = "class" ,
                    control = rpart.control(minsplit = 10 , cp = 0.01))

# Predict on test set
pred_class <- predict(tree_model , newdata = test_data , type = "class")

# Confusion matrix and accuracy

```

```

conf_mat <- confusionMatrix(pred_class, test_data$closed, positive = "1")
print(conf_mat)

# Plot the tree
rpart.plot(tree_model,
            type = 2,
            extra = 106,
            fallen.leaves = TRUE,
            box.palette = "Blues")

# Remove target column and convert predictors to matrix
X_train <- model.matrix(closed ~ . - 1, data = train_data)
y_train <- as.numeric(as.character(train_data$closed))

X_test <- model.matrix(closed ~ . - 1, data = test_data)
y_test <- as.numeric(as.character(test_data$closed))

train_data$closed <- as.numeric(as.character(train_data$closed))

# Cross validation to fine tune parameters
hyperparam_grid <- expand.grid(
  nrounds = seq(from = 200, to = 500, by = 100),
  eta = c(0.01, 0.1),
  max_depth = c(5, 6, 7),
  gamma = c(0, 1),
  colsample_bytree = c(0.5, 0.75, 1.0),
  min_child_weight = 1,
  subsample = 1
)

num_cores <- parallel::detectCores() - 1
cl <- makeCluster(num_cores)
registerDoParallel(cl)

tune_control <- trainControl(
  method = "cv",
  number = 10,
  verboseIter = FALSE,
  allowParallel = TRUE
)

xgb_cv <- train(
  closed ~ .,
  data = train_data,
  method = "xgbTree",
  trControl = tune_control,
  metric = "logLoss",
  tuneGrid = hyperparam_grid
)

```

```

stopCluster(cl)
registerDoSEQ()

final_model <- xgboost(data = as.matrix(X_train),
                      label = y_train,
                      booster = "gbtree",
                      objective = "binary:logistic",
                      nrounds = xgb_cv$bestTune$nrounds,
                      max_depth=xgb_cv$bestTune$max_depth,
                      colsample_bytree=xgb_cv$bestTune$colsample_bytree,
                      min_child_weight=xgb_cv$bestTune$min_child_weight,
                      subsample=xgb_cv$bestTune$subsample,
                      eta=xgb_cv$bestTune$eta,
                      gamma=xgb_cv$bestTune$gamma,
                      verbose = 0)

# Get predicted probabilities
y_prob <- predict(final_model, as.matrix(X_test), type = "response")

# Calculate ROC curve
roc_obj <- roc(test_data$closed, y_prob)
best_threshold <- coords(roc_obj, "best", best.method = "youden")$threshold

# Convert probabilities to classes using optimal threshold
pred_class <- ifelse(y_prob > best_threshold, 1, 0)

# Evaluate
confusionMatrix(factor(pred_class), factor(test_data$closed), positive = "1")

# Feature Importance
importance_matrix <- xgb.importance(colnames(X_train), model = final_model)
xgb.plot.importance(
  importance_matrix,
  rel_to_first = TRUE,
  xlab = "Relative importance")

# Convert date
boa_data$established_date <- mdy(boa_data$established_date)

# Extract year from date
boa_data$established_year <- year(boa_data$established_date)

boa_data_filtered <- boa_data %>%
  filter(established_year < 2024) %>%
  mutate(
    entry_time = pmax(0, 2014 - established_year),
    exit_time = 2024 - established_year, # same as duration
    closed = as.numeric(as.character(closed))
  )

```

```

) %>%
  filter(exit_time > entry_time)

boa_data_filtered <- boa_data_filtered %>%
  dplyr::select(closed,
                exit_time,
                combined_stat_area,
                deposits,
                branch_share,
                market_share,
                network_equity,
                deposit_change,
                nearby_boa_branches,
                total_nearby_competitor_branches,
                total_nearby_competitor_deposits,
                total_nearby_boa_deposits,
                estimate_household_median_income,
                estimate_household_mean_income,
                region_code,
                entry_time
                ) %>%
  filter(if_all(everything(), ~ !is.na(.) & !is.infinite(.)))

# Kaplan-Meier Survival Curve
surv_obj <- Surv(time = boa_data_filtered$entry_time,
                  time2 = boa_data_filtered$exit_time,
                  event = boa_data_filtered$closed)

# Kaplan-Meier
km_fit <- survfit(surv_obj ~ 1)

# Plot
plot(km_fit,
      xlab = "Year",
      ylab = "Survival Probability",
      col = "steelblue")

# Check Assumptions
zph <- cox.zph(cox_full)
print(zph)

cox_time_interact <- coxph(
  Surv(entry_time, exit_time, closed) ~
    tt(deposits) + tt(deposit_change) +
    network_equity + estimate_household_median_income + region_code,
  data = boa_data_filtered,
  tt = function(x, t, ...) x * log(t)
)
summary(cox_time_interact)

```

```

folds <- createFolds(boa_data_filtered$closed, k = 5, list = TRUE)
c_indices <- numeric(length(folds))

for (i in seq_along(folds)) {
  train_idx <- setdiff(seq_len(nrow(boa_data_filtered)), folds[[i]])
  train_data <- boa_data_filtered[train_idx, ]
  test_data <- boa_data_filtered[folds[[i]], ]

  # Fit time-dependent Cox model with left truncation
  model <- coxph(Surv(entry_time, exit_time, closed) ~
    tt(deposits) + tt(deposit_change) +
    network_equity + estimate_household_median_income + region_code,
    data = train_data,
    tt = function(x, t, ...) x * log(t))

  # Predict risk scores on test set
  pred <- predict(model, newdata = test_data, type = "lp")

  # Survival object for test set
  surv_test <- Surv(test_data$entry_time, test_data$exit_time, test_data$closed)

  # Concordance index
  c_indices[i] <- survConcordance(surv_test ~ pred)$concordance
}

# Report average C-index
mean_c_index <- mean(c_indices)
print(paste("Average C-index across folds:", round(mean_c_index, 4)))

```